Adaptive Resilient Output Feedback Control Against Unknown Deception Attacks for Nonlinear Cyber-Physical Systems

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***Abstract*—In this brief, by considering the sensors subject to deception attacks, the observer-based adaptive resilient control problem of a class of nonlinear cyber-physical systems is addressed. Because authentic system state information can be injected with counterfeit data, the impacts of unknown nonlinearity related to unknown attack weights become difficult to be compensated. To circumvent these obstacles, we construct a novel state observer using the system output after being attacked only, and the effects of unknown control gains due to deception attacks on the system are handled by introducing Nussbaum technique. A resilient output feedback control scheme is designed by utilizing the compromised state variables, guaranteeing boundedness for all signals within the closed-loop system. In the end, the efficacy of the proposed control scheme is affirmed through simulation results involving a single-link robot.**

***Index Terms*—Nonlinear cyber-physical systems, adaptive resilient control, unknown deception attacks, state observer.**

I. Introduction

IN THE context of today's times, the relevant interpretations of cyber-physical systems (CPSs) are expressed as the control systems that combine physical components with network communications and computer operations [1]. Due to CPSs require remote supervision and control design in an open network environment, operational data may be attacked by enemies. Hence, the most significant menace to CPSs security arises from cyber attacks [2]. Cyber attacks can be broadly classified into three categories as follows: 1) Denial of Service (DoS) attacks [3], [4], which attempt to interrupt or stop system services, making it impossible to transmit information; 2) replay attacks [5], a common type of network attacks, involve the malicious or fraudulent repetition or delay of valid data transmissions for a certain period; 3) deception attacks [6], [7], in which attackers maliciously inject false information into sensors, resulting in incomplete and inaccurate data. All three attacks previously mentioned have a non-negligible impact on the normal operation of the system, and how to ensure the stability of the system after suffering network attacks is a pressing issue.

In response to deliberate deception attacks, some research works on adaptive control schemes have been carried out methodically over the last few years. In the study by [8], an adaptive control scheme designed for linear dynamical systems when countering sensor uncertainty and sensor attacks was presented. Furthermore, [9] introduced an enhanced adaptive resilient control scheme to mitigate the effects of the network attacks, which guarantees the asymptotic stability and the performance of system. In the context of uncertain nonlinear time-delay CPSs with deception attacks, by [10], a memoryless adaptive elastic control design scheme was proposed. Due to deliberate deception attacks on CPSs, the real system state signals are not available and only the compromised states are available, which causes traditional backstepping schemes are no longer applicable. Therefore, in the above context, how to design suitable adaptive control strategy for nonlinear systems is a challenging task.

Due to the existence of unknown nonlinear terms in the models of nonlinear systems, efficiently handling these terms has become a topical research problem. With the rapid development of control engineering in recent years, the fuzzy logic systems (FLSs) and neural networks (NNs) have been extensively employed to address control challenges in uncertain nonlinear systems, yielding numerous related findings as demonstrated in [11], [12]. In [13], an adaptive control scheme employing NNs had been designed for single-input single-output (SISO) nonlinear systems. In [14], a fuzzy state observer for nonstrict feedback nonlinear systems was applied to estimate information about the unmeasurable state variables.

In this brief, the main objective is to explore the problem of designing observer-based adaptive resilient control for nonlinear CPSs under deception attacks. The main contributions are summarized below: (**i**) This brief introduces a novel state observer designed to address the issue of unavailable state variables resulting from deception attacks by utilizing only

Manuscript received 11 September 2023; revised 16 December 2023; accepted 27 February 2024. Date of publication 4 March 2024; date of current version 31 July 2024. This work was supported by the National Natural Science Foundation of China under Grant 62303287. This brief was recommended by Associate Editor C. Wang. *(Corresponding author: Xinjun Wang.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCSII.2024.3372413.

Digital Object Identifier 10.1109/TCSII.2024.3372413

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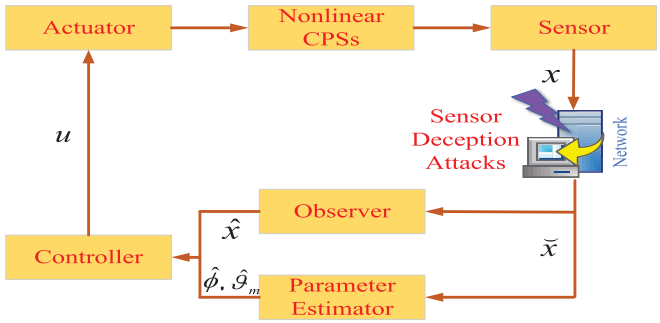


Fig. 1. The control framework of nonlinear CPS (1) under sensor deception attacks.

the attacked output to estimate the unmeasured system states, which can facilitate the implementation of output feedback control. (**ii**) We introduce a novel coordinate transformation that incorporates the attacked output and estimated states. This transformation enables the development of backstepping design approach for achieving steady-state performance under deception attacks. Additionally, it ensures that error signals can converge to a small neighborhood around the origin. (**iii**) Compared with the existing result [15], the singularity issues arising in the adaptive iterative process are avoided by introducing the auxiliary variables in this brief. Furthermore, the aid of Nussbaum technique is effectively employed to address the impact of unknown time-varying gains resulting from the attacks.

II. Problem Formulation and Technical Tools

A. Description of the System

This brief studies the nonlinear CPSs under deception attacks through sensor networks, which are described as follows:

$$\begin{cases} \dot{x}_k = f_k(\bar{x}_k) + x_{k+1}, \\ \dot{x}_n = f_n(x_n) + u, \\ \check{x}_k = \eth(t, x_k) + x_k, \\ y = x_1, \end{cases} \tag{1}$$

where $k = 1, \ldots, n$, $\bar{x}_k = [x_1, x_2, \ldots, x_k]^T$, $u \in \mathcal{R}$ and $y \in \mathcal{R}$ are the state vectors, input and output of the system, respectively. $\check{x}_k$ are the available compromised system states after attacks. $f_k(\cdot)$ denote the unknown and smooth functions, $\eth(t, x_k)$ represent the false data injection attacks. The control framework of nonlinear CPS (1) under unknown sensor deception attacks is described in Fig. 1.

This brief considers the nonlinear CPS (1) under sensor deception attacks and the main goal is to design an adaptive resilient control scheme which ensures the boundedness of all the resulting closed-loop system signals.

Assumption 1: In this brief, it is assumed that the attacks are state-dependent and parameterized as $\eth(t, x_k(t)) = \hbar(t)x_k(t)$, where $\hbar(t)$ represent the unknown time-varying weights, and there exist unknown constants $\bar{\hbar}$ and $\hbar^*$ satisfy $|\hbar(t)| \leq \bar{\hbar}$ and $|\dot{\hbar}(t)| \leq \hbar^*$, respectively. In addition, $\hbar(t)$ are subject to the following conditions: $1 + \hbar(t) \neq 0$.

Definition 1 [15]: A Nussbaum-type function $\mathbb{N}(s)$, which is a continuous differentiable function on $(0, \infty] \Rrightarrow (-\infty, +\infty)$, satisfies the following equations: $\lim_{s\to\pm\infty} sup \frac{1}{s}\int_0^s \mathbb{N}(y)dy = +\infty$, $\lim_{s\to\pm\infty} inf \frac{1}{s}\int_0^s \mathbb{N}(y)dy = -\infty$.

Lemma 1 [15]: The function $\mathbb{N}(\divideontimes)$ represents a Nussbaum-type function, while two smooth functions $\mathcal{V}(\divideontimes)$ and $\mu(\divideontimes)$, defined on the interval $[0, \infty)$ satisfy $\mathcal{V}(t) \geq 0$. The time-varying function $\Bbbk(t)$ satisfies the condition $\Bbbk(t) \in \mho = [\Bbbk^-, \Bbbk^+]$ (the constants $\Bbbk^+$ and $\Bbbk^-$ must satisfy $\Bbbk^-\Bbbk^+ > 0$), of which $0 \notin \mho$. Suppose that there are two positive constants $\mathcal{C}$ and $\mathcal{D}$, the following inequality holds $\dot{\mathcal{V}}(t) \leq -\mathcal{C}\mathcal{V} + \sum_{m=1}^{n}(\Bbbk(t)\mathbb{N}(\mu(t)) + 1)\dot{\mu}(t) + \mathcal{D}$, where the condition $\forall t > 0$, $\dot{\mu}(t) \geq 0$. $\mathcal{V}(t)$ and $\mu(t)$ remain bounded over the interval $[0, \infty)$.

B. Neural Networks Technology

This brief employs radial basis function neural networks (RBFNNs) for approximating the unknown continuous nonlinear terms within the system. By the universal approximation property of RBFNNs as demonstrated in [16], RBFNNs can approximate $f_q(Z_q) : R^q \rightarrowtail R$ over a compact set $\Omega_{Z_q} \subset R^q$ as below: $f_q(Z_q) = w_q^{*T} S_q(Z_q) + \epsilon_q(Z_q)$, where $q = 1, \ldots, n$, $\epsilon_q(Z_q)$ denote the approximation errors, $w_q^* = \arg\min_{w_q \in R^{l_q}} \{ \sup_{Z_q \in \Omega_{Z_q}} |f_q(Z_q) - w_q^T S_q(Z_q)| \}$ with $Z_q \in \Omega_{Z_q}$ representing the input vectors. $w_q = [\delta_1, \delta_2, \ldots, \delta_{l_q}]^T$ denote the weight vectors and $l_q > 1$ are the node numbers. $S_q(Z_q) = [s_{q,1}(Z_q), \ldots, s_{q,l_q}(Z_q)]^T \in R^{l_q}$ denote the basis function vectors. Generally, the Gaussian functions are chosen as the basis functions as follows:

$$S_{q,j}(Z_q) = \exp\Big[\frac{-(Z_q - \widetilde{c}_{q,j})^T (Z_q - \widetilde{c}_{q,j})}{2\widetilde{b}_{q,j}^2}\Big], \tag{2}$$

where $j = 1, 2, \ldots, l_q$, $\widetilde{b}_{q,j}$ represent the widths, and $\widetilde{c}_{q,j}$ represent the center of the Gaussian functions.

Lemma 2 [17]: By the basis function vectors expressed in (2), they satisfy $\|S_q(Z_q)\|^2 \leq \ell$, in which ℓ is the node number.

Lemma 3 [18]: For the sets $\bar{x}_q = [x_1, \ldots, x_k, \ldots, x_q]^T$ and $\bar{x}_k = [x_1, \ldots, x_k]^T$. For the description of the basis function vectors in (2), it can be expressed as $S(\bar{x}_k) = [S_1(\bar{x}_k), \ldots, S_\ell(\bar{x}_k)] \in R^\ell$ and $S(\bar{x}_q) = [S_1(\bar{x}_q), \ldots, S_\ell(\bar{x}_q)] \in R^\ell$. Then, for $\forall k$, $\forall q$ satisfy k and $q \in N^+$ with $k \leq q$, the inequality $\|S(\bar{x}_q)\|^2 \leq \|S(\bar{x}_k)\|^2$ holds.

Assumption 2: The weight vectors w_q and errors $\epsilon_q(Z_q)$ are unknown and bounded, which satisfy $\|w_q\| \leq \bar{w}$ and $|\epsilon_q(Z_q)| \leq \bar{\epsilon}$ with $\bar{w}$ and $\bar{\epsilon}$ being unknown positive constants, respectively.

III. Main Results

A. Design of a Novel State Observer

For the nonlinear system (1), the sensors are maliciously injected with false data, then, the precise state variables information from the system cannot be measured, only the compromised states $\check{x}_i$ can be obtained. Therefore, a novel state

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observer is proposed to deal with this difficulty, which can be defined in the following form:

$$\begin{cases} \dot{\hat{x}} = \mathcal{A}\hat{x} + B_n u + \mathcal{L}\check{x}_1, \\ \hat{y} = \hat{x}_1, \end{cases} \tag{3}$$

where $\hat{x} = [\hat{x}_1, \ldots, \hat{x}_n]^T$, $\hat{x}_i = [\hat{x}_1, \ldots, \hat{x}_i]^T$ are the estimate of $\bar{x}_i = [x_1, \ldots, x_i]^T$, $\mathcal{L} = \left[l_1, l_2, \ldots, l_n\right]^T$, $B_n = [0, \ldots, n]^T$ $\mathcal{A} = \begin{bmatrix} -l_1 & & \\ \vdots & I_{n-1} & \\ -l_n & \cdots & 0 \end{bmatrix}$ with $\mathcal{L}$ being the design matrix such that the matrix $\mathcal{A}$ is a strict Hurwitz matrix. There must exist symmetric $\mathcal{P} > 0$, $\mathcal{Q} > 0$ satisfying $\mathcal{A}^T\mathcal{P} + \mathcal{P}\mathcal{A} = -2\mathcal{Q}$.

Then, let $\varrho = x - \hat{x} = [\varrho_1, \varrho_2, \ldots, \varrho_n]^T$ be an estimate error, the following equation holds:

$$\dot{\varrho} = \mathcal{A}\varrho + \mathcal{F}(\bar{x}) + \mathcal{L}\eth(t, x_1) \tag{4}$$

with $\mathcal{F}(\bar{x}) = [f_1(\bar{x}_1), \ldots, f_n(\bar{x}_n)]^T$. Since $\mathcal{F}(\bar{x})$ contain the unknown functions, so we employ RBFNNs to handle these functions, which satisfy $\mathcal{F}(\bar{x}) = w_0^T S_0(X_0) + \epsilon_0(X_0)$, where $X_0 = \bar{x} \in \Omega_0$ (Ω_0 is a compact set for x) and $w_0 = (w_{10}^T, w_{20}^T, \ldots, w_{n0}^T)$, the error $\epsilon_0(X_0) = [\epsilon_{10}(X_0), \epsilon_{20}(X_0), \ldots, \epsilon_{n0}(X_0)]^T$ with $\|\epsilon_0(X_0)\| \le \bar{\epsilon}_0$.

The estimation error ϱ can be rewritten as

$$\dot{\varrho} = \mathcal{A}\varrho + w_0^T S_0(X_0) + \epsilon_0(X_0) + \mathcal{L}\eth(t, x_1). \tag{5}$$

Define the Lyapunov function $V_0 = \frac{1}{2}\varrho^T\mathcal{P}\varrho$. V_0 can be derived as

$$\begin{aligned} \dot{V}_0 = {} & \frac{1}{2}\varrho^T(\mathcal{P}\mathcal{A} + \mathcal{A}^T\mathcal{P})\varrho + \varrho^T\mathcal{P}(w_0^T S_0(X_0) + \epsilon_0(X_0)) \\ & + \varrho^T\mathcal{P}\mathcal{L}\eth(t, x_1). \end{aligned} \tag{6}$$

Utilizing Young's inequality, we have

$$\begin{aligned} & \varrho^T\mathcal{P}(w_0^T S_0(X_0) + \epsilon_0(X_0)) \le \|\mathcal{P}\|^2\bar{\epsilon}_0^2 + \|\mathcal{P}\|^2\vartheta_m + \|\varrho\|^2, \\ & \varrho^T\mathcal{P}\mathcal{L}\eth(t, x_1) \le \frac{1}{2}\|\varrho\|^2 + \frac{1}{2}\|\mathcal{P}\|^2\|\mathcal{L}\|^2\bar{\eth}_1^2, \end{aligned} \tag{7}$$

where $|\eth(t, x_1)| \le \bar{\eth}_1$. For the sake of subsequent calculations, we introduce a constant ϑ_m that appears in the subsequent calculations, it takes the following form: $\vartheta_m = \max\{\delta_j : j = 1, 2, \ldots, n\}$, where δ_j are unknown constants that will be defined in the subsequent contents, $\tilde{\vartheta}_m$ is the defined estimation error of the form $\tilde{\vartheta}_m = \vartheta_m - \hat{\vartheta}_m$, and $\hat{\vartheta}_m$ is the estimate of ϑ_m.

After the above analysis, we can easily obtain the following inequality:

$$\begin{aligned} \dot{V}_0 \le {} & -\Delta\|\varrho\|^2 + \|\mathcal{P}\|^2\bar{\epsilon}_0^2 + \|\mathcal{P}\|^2\vartheta_m \\ & + \frac{1}{2}\|\mathcal{P}\|^2\|\mathcal{L}\|^2\bar{\eth}_1^2, \end{aligned} \tag{8}$$

where $\Delta = (\lambda_{min}(\mathcal{Q}) - \frac{3}{2})$.

Remark 1: Due to the boundedness of ϑ_m is unknown, it is important to note that stability of the state observer (3) cannot be directly inferred from (8). But, in the next section, the boundedness of ϑ_m can be realized by employing adaptive control techniques. Then, it can be seen from (8) that if Δ is chosen to be larger enough and $\bar{\epsilon}_0$, ϑ_m and $\bar{\eth}_1$ are bounded, we can deduce that the observer estimation error is bounded, validating the soundness of the state observer design in this brief.

B. Recursive Adaptive Resilient Control Design

This section mainly discusses the controller design of the nonlinear system (1) after suffering from the deception attacks. This is a huge challenge for the controller design as the real state variables $x_k (k = 1, 2, \ldots, n)$ are not available due to the injection of false data.

Thus, we introduce a coordinate transformation as shown below

$$\zeta_1 = x_1, \zeta_k = \lambda\hat{x}_k - \lambda\alpha_{k-1}, k = 2, \ldots, n, \tag{9}$$

which implies

$$\check{\zeta}_1 = \check{x}_1, \check{\zeta}_k = \hat{x}_k - \alpha_{k-1}, \quad k = 2, \ldots, n, \tag{10}$$

where $k = 2, 3, \ldots, n$, α_{k-1} are the designed virtual controllers. Here, we define the unknown positive constants $\lambda_m, \lambda_M, \vec{\lambda}$. According to ***Assumption 1*** and $\lambda = (1 + \hbar)^{-1}$, we can get $0 < \lambda_m \le |\lambda| \le \lambda_M$ and $|\dot{\lambda}| \le \vec{\lambda}$. Based on the above coordinate transformation (9), (10), we can easily get $x_k = \lambda\check{x}_k$ and $\zeta_k = \lambda\check{\zeta}_k$ hold.

***Step 1*:** Consider the Lyapunov function $V_1 = V_0 + \frac{\lambda^2}{2\lambda_M}\zeta_1^2 + \frac{1}{2}\tilde{\kappa}^2 + \frac{1}{2}\tilde{\vartheta}_m^2$, where $\tilde{\kappa} = \kappa - \hat{\kappa}$ is the estimation error. The estimate of κ is denoted as $\hat{\kappa}$ with $\kappa = \lambda_M^2\vec{\lambda}$. Then

$$\begin{aligned} \dot{V}_1 \le {} & \dot{V}_0 + \kappa\check{\zeta}_1^2 + \frac{\zeta_1^2 + \zeta_2^2}{2} + \tilde{\vartheta}_m\dot{\tilde{\vartheta}}_m + \tilde{\kappa}\dot{\tilde{\kappa}} \\ & + \frac{\lambda^2}{\lambda_M}\zeta_1(\alpha_1 + \bar{f}_1), \end{aligned} \tag{11}$$

where $\bar{f}_1 = f_1 + \varrho_2$, which satisfies $\bar{f}_1 = w_1^T S_1(\vec{\Xi}_1) + \epsilon_1$ with $\vec{\Xi}_1 = [x_1, x_2]^T$. ϵ_1 is the approximation error and $|\epsilon_1| \le \bar{\epsilon}_1$. Combining ***Lemma 3*** and the Young's inequality, the following inequality is established: $\frac{\lambda^2}{\lambda_M}\zeta_1\bar{f}_1 \le \frac{1}{2}\check{\zeta}_1^2\vartheta_m S_1^T(\Xi_1)S_1(\Xi_1) + \frac{\lambda_M^2\bar{\epsilon}_1^2}{2} + \frac{\zeta_1^2}{2} + \frac{1}{2}$, where $\Xi_1 = [x_1]^T$, $\delta_1 = \lambda_M^4\|w_1\|^2$.

Further, we can get the following formula:

$$\begin{aligned} \dot{V}_1 \le {} & \dot{V}_0 + \zeta_1^2 + \frac{\zeta_2^2}{2} + \frac{\lambda^2}{\lambda_M}\zeta_1\alpha_1 + \check{\zeta}_1^2\aleph_1 + \tilde{\vartheta}_m\dot{\tilde{\vartheta}}_m + \tilde{\kappa}\dot{\tilde{\kappa}} \\ & + \frac{1}{2}\check{\zeta}_1^2\vartheta_m S_1^T(\check{\Xi}_1)S_1(\check{\Xi}_1) + \frac{\lambda_M^2\bar{\epsilon}_1^2}{2} + \frac{1}{2}, \end{aligned} \tag{12}$$

where $\aleph_1 = \frac{1}{2}\vartheta_m(S_1^T(\Xi_1)S_1(\Xi_1) - S_1^T(\check{\Xi}_1)S_1(\check{\Xi}_1))$ with $\check{\Xi}_1 = [\check{x}_1]^T$.

The virtual controller α_1 is designed as

$$\alpha_1 = -k_1\check{\zeta}_1 - \check{\zeta}_1 + N(\mu_1)\Phi_1, \tag{13}$$

where $\Phi_1 = \frac{1}{2}\check{\zeta}_1^3 + \check{\zeta}_1\hat{\kappa} + \frac{1}{2}\check{\zeta}_1\hat{\vartheta}_m S_1^T(\check{\Xi}_1)S_1(\check{\Xi}_1)$, $\dot{\mu}_1 = \check{\zeta}_1\Phi_1$, $\dot{\hat{\kappa}} = \check{\zeta}_1^2 - r\hat{\kappa}$ with k_1 and r being positive constants to be designed.

After the above derivation, we can obtain

$$\begin{aligned} \dot{V}_1 \le {} & -k_1\zeta_1^2 + \dot{V}_0 + \frac{1}{2}\zeta_2^2 + (g_1(t)N(\mu_1) + 1)\dot{\mu}_1 \\ & + \frac{1}{2}\aleph_1^2 + \tilde{\vartheta}_m(\pi_1 - \dot{\hat{\vartheta}}_m) + k_0\hat{\vartheta}_m\tilde{\vartheta}_m + r\hat{\kappa}\tilde{\kappa} \\ & + \frac{\lambda_M^2\bar{\epsilon}_1^2}{2} + \frac{1}{2}, \end{aligned} \tag{14}$$

where $g_1(t) = \lambda^3/\lambda_M$, $\pi_1 = \sigma_1 - k_0\hat{\vartheta}_m$, $\sigma_1 = \frac{1}{2}\check{\zeta}_1^2 S_1^T(\check{\Xi}_1)S_1(\check{\Xi}_1)$ with π_1 being an auxiliary variable.

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Step j $(2 \le j \le n)$**:** The Lyapunov function V_2 is designed as $V_2 = V_1 + \frac{1}{2}\zeta_2^2 + \frac{1}{2}\tilde{\phi}^2$ and $V_j(3 \le j \le n)$ is designed as

$$V_j = V_{j-1} + \frac{1}{2}\zeta_j^2, \tag{15}$$

where $\tilde{\phi} = \phi - \hat{\phi}$ is the estimation error. The estimate of ϕ is denoted as $\hat{\phi}$ with $\phi = \vec{\lambda}^2\lambda_M^2$.

Defining $C_j = \sum_{i=1}^{j-1}(\partial\alpha_{j-1}/\partial\check{x}_i)\check{x}_i - \alpha_{j-1} + \hat{x}_j$, $\zeta_j\dot{\lambda}C_j \le (1/2)\check{\zeta}_j^2\vec{\lambda}^2\lambda_M^2C_j^2 + 1/2$. $\zeta_j\sum_{i=1}^{j-1}(\partial\alpha_{j-1}/\partial\check{x}_i)f_i \le (1/2)\check{\zeta}_j^2\vartheta_m S_j^T(\Xi_j)S_j(\Xi_j) + (1/2) + (\zeta_j^2/2) + (\bar{\epsilon}_j^2/2)$ with $\Xi_j = [\bar{x}_j]^T$, $\delta_j = \lambda_M^2\|w_j\|^2$, and ϵ_j is the approximation error which satisfies $|\epsilon_j| \le \bar{\epsilon}_j$.

Then, the virtual controller α_2 is designed as $\alpha_2 = -k_2\check{\zeta}_2 - \frac{3}{2}\check{\zeta}_2 + \frac{\partial\alpha_1}{\partial\check{x}_1}\check{x}_2 + \frac{\partial\alpha_1}{\partial\hat{\vartheta}_m}\pi_2 + \frac{\partial\alpha_1}{\partial\hat{\kappa}}\dot{\hat{\kappa}} + \frac{\partial\alpha_1}{\partial\mu_1}\dot{\mu}_1 - l_2(\check{x}_1 - \hat{x}_1) + N(\mu_2)\Phi_2$ and $\alpha_j(3 \le j \le n)$ is chosen as

$$\begin{aligned}\alpha_j = & -k_j\check{\zeta}_j - \frac{3}{2}\check{\zeta}_j - l_j(\check{x}_1 - \hat{x}_1) + N(\mu_j)\Phi_j + \frac{\partial\alpha_{j-1}}{\partial\hat{\vartheta}_m}\pi_j \\ & + \sum_{i=1}^{j-1}\frac{\partial\alpha_{j-1}}{\partial\check{x}_i}\check{x}_{i+1} + \sum_{i=2}^{j-1}\check{\zeta}_i\frac{\partial\alpha_{i-1}}{\partial\hat{\vartheta}_m}\sigma_j + \sum_{i=1}^{j-1}\frac{\partial\alpha_{j-1}}{\partial\mu_i}\dot{\mu}_i \\ & + \sum_{i=3}^{j-1}\check{\zeta}_i\frac{\partial\alpha_i}{\partial\hat{\iota}}\eta_{j-1} + \frac{\partial\alpha_{j-1}}{\partial\hat{\phi}}\iota_{j-1} + \frac{\partial\alpha_{j-1}}{\partial\hat{\kappa}}\dot{\hat{\kappa}},\end{aligned} \tag{16}$$

where $\Phi_j = \frac{1}{2}\check{\zeta}_j^3 + \frac{1}{2}\check{\zeta}_j\hat{\phi}C_j^2 + \frac{1}{2}\check{\zeta}_j\hat{\vartheta}_m S_j^T(\check{\Xi}_j)S_j(\check{\Xi}_j)$, $\dot{\mu}_j = \check{\zeta}_j\Phi_j$. Specially, when $j = n$, the controller u is designed to be $u = \alpha_n - \check{\zeta}_n$.

Substituting (16) into (15) yields

$$\begin{aligned}\dot{V}_j \le & \dot{V}_0 - \sum_{i=1}^{j}k_i\zeta_i^2 + \frac{1}{2}\zeta_{j+1}^2 + \sum_{i=1}^{j}(g_i(t)N(\mu_j) + 1)\dot{\mu}_j \\ & - \lambda\sum_{i=3}^{j}\zeta_i\frac{\partial\alpha_i}{\partial\hat{\kappa}}(\dot{\hat{\kappa}} - \iota_{j-1}) - \lambda\sum_{i=2}^{j}\zeta_i\frac{\partial\alpha_i}{\partial\hat{\vartheta}_m}(\dot{\hat{\vartheta}}_m - \pi_j) \\ & + \tilde{\vartheta}_m(\pi_j - \dot{\hat{\vartheta}}_m) + \tilde{\phi}(\iota_{j-1} - \dot{\hat{\phi}}) + r\hat{\kappa}\tilde{\kappa} + k_0\hat{\vartheta}_m\tilde{\vartheta}_m \\ & + \omega\hat{\phi}\tilde{\phi} + \sum_{i=1}^{j}\frac{\aleph_i^2}{2} + s_j + \frac{2j-1}{2},\end{aligned} \tag{17}$$

where $\pi_j = \check{\zeta}_j\sigma_j + \pi_{j-1}$, $\sigma_j = \frac{1}{2}\check{\zeta}_j^2S_j^T(\check{\Xi}_j)S_j(\check{\Xi}_j)$, $\iota_{j-1} = \check{\zeta}_j\eta_{j-1} + \iota_{j-2}$, $\eta_{j-1} = \frac{1}{2}\check{\zeta}_j^2C_j^2$, $C_j = \alpha_{j-1} - \sum_{i=1}^{j-1}(\frac{\partial\alpha_{j-1}}{\partial\check{x}_i}\check{x}_i) + \hat{x}_j$, $s_j = \frac{\lambda_M^2\bar{\epsilon}_1^2}{2} + \sum_{i=2}^{j}\frac{\bar{\epsilon}_i^2}{2}$, $g_j(t) = \lambda^2$, $\aleph_j = \frac{1}{2}\vartheta_m(S_j^T(\Xi_j)S_j(\Xi_j) - S_j^T(\check{\Xi}_j)S_j(\check{\Xi}_j))$ with ω being positive constant to be designed, π_j and ι_{j-1} being the auxiliary variables and $\check{\Xi}_j = [\check{\bar{x}}_j]^T$. In the ***Step n***, we design the adaptive laws $\dot{\hat{\phi}}$ and $\dot{\hat{\vartheta}}_m$ to be $\dot{\hat{\phi}} = \iota_{n-1}$, $\dot{\hat{\vartheta}}_m = \pi_n$.

To analyze the stability of the closed-loop system (1) under deception attacks, we select the total Lyapunov function V as $V = V_n$.

Theorem 1: Given the nonlinear CPS (1) under unknown malicious deception attacks under ***Assumptions 1-2***, which include the designed controller u and the selected parameter laws $\dot{\hat{\phi}}$ and $\dot{\hat{\vartheta}}_m$, then it can concluded that all signals in the closed-loop system and the observer estimation errors are ensured to be bounded by adjusting suitable parameters.

Proof: From the total Lyapunov function and (17), we know

$$\begin{aligned}\dot{V} \le & \dot{V}_0 - \sum_{i=1}^{n}k_i\zeta_i^2 + \sum_{i=1}^{n}(g_i(t)N(\mu_i) + 1)\dot{\mu}_i \\ & + \sum_{i=1}^{n}\frac{\aleph_i^2}{2} + s_n + \omega\hat{\phi}\tilde{\phi} + r\hat{\kappa}\tilde{\kappa} + k_0\hat{\vartheta}_m\tilde{\vartheta}_m \\ & + \frac{2n-1}{2}.\end{aligned} \tag{18}$$

By means of ***Assumption 2*** and ***Lemma 2***, we can learn that the constant $\aleph_j$ is bounded and satisfies $|\aleph_j| \le \bar{\aleph}_j$ with $\bar{\aleph}_j > 0$.

From $\omega\hat{\phi}\tilde{\phi} \le -\frac{\omega}{2}\tilde{\phi}^2 + \frac{\omega}{2}\phi^2$, $r\hat{\kappa}\tilde{\kappa} \le -\frac{r}{2}\tilde{\kappa}^2 + \frac{r}{2}\kappa^2$, $k_0\hat{\vartheta}_m\tilde{\vartheta}_m \le -\frac{k_0}{2}\tilde{\vartheta}_m^2 + \frac{k_0}{2}\vartheta_m^2$, we can obtain

$$\begin{aligned}\dot{V} \le & \dot{V}_0 - \sum_{i=1}^{n}k_i\zeta_i^2 + \sum_{i=1}^{n}(g_i(t)N(\mu_i) + 1)\dot{\mu}_i \\ & + \sum_{i=1}^{n}\frac{\bar{\aleph}_i^2}{2} + \frac{\omega}{2}\phi^2 + \frac{r}{2}\kappa^2 + \frac{k_0}{2}\vartheta_m^2 + s_n \\ & - \frac{\omega}{2}\tilde{\phi}^2 - \frac{r}{2}\tilde{\kappa}^2 - \frac{k_0}{2}\tilde{\vartheta}_m^2 + \frac{2n-1}{2}.\end{aligned} \tag{19}$$

By previous analysis, we define

$$\begin{aligned}h_{min} = & \min\Big\{\frac{\Delta}{\lambda_{max}(\mathcal{P})}, \frac{2k_1\lambda_M}{\lambda^2}, 2k_2, \ldots, 2k_n, k_0, \omega, r\Big\}, \\ \chi = & \sum_{i=1}^{n}\frac{\bar{\aleph}_i^2}{2} + \frac{k_0}{2}\vartheta_m^2 + \frac{\omega}{2}\phi^2 + \frac{r}{2}\kappa^2 + \|\mathcal{P}\|^2\|\mathcal{L}\|^2\bar{\Theta}_1^2 \\ & + \|\mathcal{P}\|^2\vartheta_m + \|\mathcal{P}\|^2\bar{\epsilon}_0^2 + s_n + \frac{2n-1}{2}.\end{aligned} \tag{20}$$

Thereby, the following inequality holds:

$$\begin{aligned}V(t) \le & C_0e^{-h_{min}t}\int_0^t\sum_{i=1}^{n}(g_i(t)N(\mu_i) + 1)\dot{\mu}_ie^{h_{min}\mu}d\mu \\ & + \mathcal{G},\end{aligned} \tag{21}$$

where $\mathcal{G} = V(0) + \frac{\chi}{h_{min}}$.

According to (21), it is known that all signals within the closed-loop system are proven to be bounded. In addition, from the designed parameter adaptive laws, the observer estimation errors are bounded. ■

Remark 2: In the above proof process, it is underscores the profound connection between the stability of the closed-loop system and the selected parameters. In order to ensure the stability of the closed-loop system, the associated positive parameters $k_i(i = 1, \ldots, n)$ are set to fairly large values, while ω, r and k_0 are minimized. However, identifying an optimal set that ensures the desired performance remains an ongoing challenge. In the simulation, a trial-and-error approach is applied.

Remark 3: In the step i $(3 \le i \le n)$ of [15], the items $1/\check{z}_i\sum_{p=1}^{i-2}\sum_{m=p+1}^{i-1}\check{z}_m[\partial\alpha_{m-1}/\partial\hat{\lambda}_p]\omega_{p,i-p+1}$ were generated in the process of designing the virtual and actual controllers, where $1/\check{z}_i$ may lead to the creation of singularity. In view of this, we solve this difficulty by constructing a new auxiliary variable: $\pi_j = \check{\zeta}_j\sigma_j + \pi_{j-1}$. The proposed control scheme in this brief helps circumvent the singularity issue during the adaptive iterative process.

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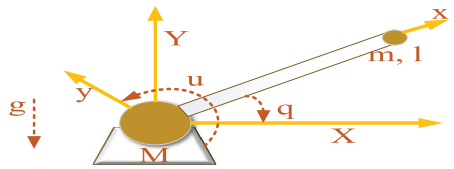


Fig. 2. Framework of single-link robot.

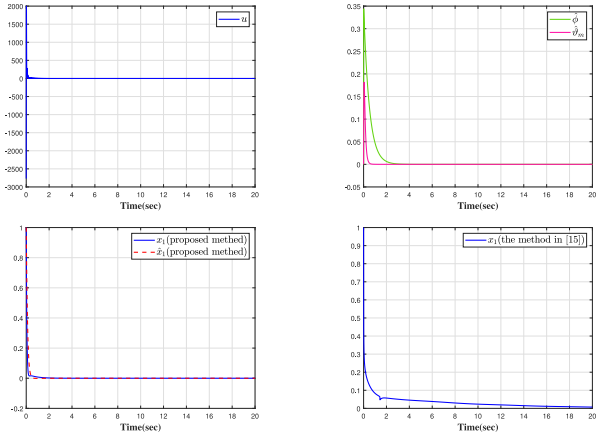


Fig. 3. The performance of the system under deception attacks.

IV. Simulation Studies

In this brief, we employ the single-link robot shown in Fig. 2 to demonstrate the applicability of the proposed control approach [19]: $2M\ddot{q} + mgl\sin(q) = 2u, y = q$, where q is the horizontal angle of the mechanical rod to the ground. $M = 2$ is the rotational inertia of the connection, the input torque is u, the acceleration of gravity g is taken as 9.8. The mass m and length l of the arm are both taken as 1.

In the simulation, we initialize the states with the following initial values: $x_1(0) = 1$ and $x_2(0) = 0$. The attacks are modeled using $\hbar(t) = -1 + 0.05\cos(t)$. The resulting parameters are designed as $k_0 = 10$, $k_1 = 10$. $k_2 = 1101$ and $N_i(\mu_i) = 2\mu_i e^{\mu_i^2}\sin(2^i\mu_i) + e^{\mu_i^2}2^i\cos(2^i\mu_i)$, $i = 1, 2$.

We present the simulation results in Fig. 3, illustrating the control input of system, the trajectories of the adaptive laws $\hat{\phi}$ and $\hat{\vartheta}_m$. In order to illustrate the advantages of the proposed control method under deception attacks we compared it with a representative control scheme proposed in [15]. Utilizing identical initial conditions and design parameters, simulations were conducted with a time step of 0.01s. The figure shows the respective time required for output convergence. In comparison with the approach presented in [15], our proposed control method not only achieves similar performance, but also greatly reduces the time required for output convergence.

V. Conclusion

This brief has investigated the strategy of an adaptive resilient output feedback control for nonlinear CPSs under sensor deception attacks design. An observer-based adaptive resilient control scheme has been formulated by introducing a novel state observer and incorporating the backstepping design approach. The proposed control scheme ensures the boundedness of all signals in the closed-loop system, including the observer estimation error. In the engineering context, many real dynamic systems are constructed as non-integer order dynamic models based on fractional-order calculus [20]. Therefore, our future research direction is focused on achieving fractional order control under deception attacks.

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